WORKED SOLUTION

# STEP II 2013, Question 3

Pure • Roots of Cubics

**STEP II 2013, Question 3**

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/13-s2-q3

## The key idea

Everything follows from the shape of a cubic with positive leading coefficient and three distinct real roots: it rises from $-\infty$, crosses the axis upwards, comes back down, then crosses upwards again. The value $f(0) = c$ tells us where 0 sits relative to the roots, and the turning points (the roots of a quadratic, $f'(x) = 0$) can be pinned down by their sum and product. The last part is a reminder that conditions on the turning points alone cannot guarantee that the curve actually reaches the axis three times.

## Solution

Throughout, let

$$f(x) = x^3 + 3ax^2 + 3bx + c, \qquad f'(x) = 3(x^2 + 2ax + b).$$

When $f(x) = 0$ has three distinct real roots $\alpha < \beta < \gamma$, we can write $f(x) = (x-\alpha)(x-\beta)(x-\gamma)$, so, reading off the signs of the three factors,

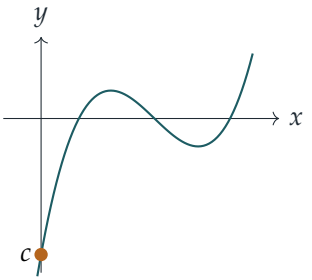
$$f(x) < 0 \text{ on } (-\infty, \alpha) \cup (\beta, \gamma), \qquad f(x) > 0 \text{ on } (\alpha, \beta) \cup (\gamma, \infty).$$

Also, by Rolle's theorem $f'$ has a zero in $(\alpha, \beta)$ and another in $(\beta, \gamma)$. Since $f'$ is a quadratic it has no other zeros, so the turning points are at $p < q$ with

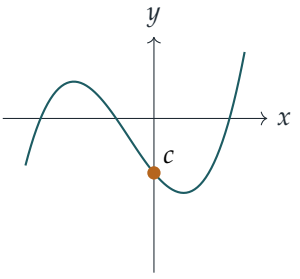
$$\alpha < p < \beta < q < \gamma.$$

### Part (i)

The curve $y = f(x)$ meets the $y$-axis at $(0, c)$, which is below the $x$-axis since $c < 0$. From the sign pattern above, $f(0) < 0$ means that either $0 < \alpha$ or $\beta < 0 < \gamma$. These are the two pictures below.



all three roots positive



exactly one root positive

If $0 < \alpha$, all three roots are positive. If $\beta < 0 < \gamma$, then $\alpha < \beta < 0$, so exactly one root ($\gamma$) is positive. No root is zero, because $f(0) = c \neq 0$. So either exactly one root is positive or all three are, as required.

**Part (ii)**

Now $0 < \alpha < \beta < \gamma$.

*The sign of c.* Since $0 < \alpha$, the point 0 lies in $(-\infty, \alpha)$, where $f < 0$. So $c = f(0) < 0$.

*The turning points.* The equation $x^2 + 2ax + b = 0$ has the two distinct real roots $p$ and $q$, so its discriminant is positive:

$$4a^2 - 4b > 0 \quad \Longrightarrow \quad a^2 > b.$$

Both turning points are positive, because $p > \alpha > 0$ and $q > p$. For the quadratic $x^2 + 2ax + b$, the sum and product of the roots are

$$p + q = -2a, \qquad pq = b.$$

Since $p, q > 0$, we get $-2a > 0$, so $a < 0$, and $b = pq > 0$. Putting these together,

$$a^2 > b > 0, \qquad a < 0, \qquad c < 0,$$

which is $(*)$.

(You can also get $a < 0$, $b > 0$ and $c < 0$ straight from the roots: $\alpha + \beta + \gamma = -3a$, $\alpha\beta + \beta\gamma + \gamma\alpha = 3b$ and $\alpha\beta\gamma = -c$ are all positive. But $a^2 > b$ really does need the turning points.)

**Part (iii)**

Now $c = f(0) > 0$, so 0 lies in $(\alpha, \beta)$ or in $(\gamma, \infty)$, the two intervals where $f > 0$. That gives two candidate pictures: either $\alpha < 0 < \beta < \gamma$ (one negative root, two positive), or $\alpha < \beta < \gamma < 0$ (all three negative).

**?** **Ask yourself** How can the condition $ab < 0$ rule one of these out?

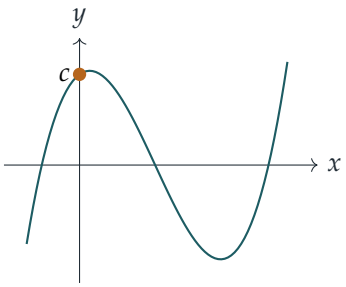
Look at the turning points again. If all three roots were negative, then $p < q < \gamma < 0$, so both turning points would be negative. Then

$$-2a = p + q < 0 \quad \text{and} \quad b = pq > 0,$$

so $a > 0$ and $b > 0$, giving $ab > 0$. This contradicts $ab < 0$. Hence the second picture is impossible, and

$$\alpha < 0 < \beta < \gamma :$$

*one root is negative and the other two are positive.*




This case really can happen. For example, the roots $-1$, 2 and 5 give $x^3 - 6x^2 + 3x + 10$, so $a = -2$, $b = 1$ and $c = 10$, with $ab = -2 < 0$ and $c > 0$; this is the curve sketched above.

**Part (iv)**

**Ask yourself** What extra condition, beyond $(*)$, does a cubic need to have three real roots?

Condition $(*)$ only controls the turning points and the sign of $f(0)$. For three real roots the local maximum must also be above the axis. So we should choose $a$ and $b$ to give two positive turning points, and then make $c$ negative enough to pull the local maximum below the axis.

Take $a = -1$ and $b = \frac{3}{4}$, so that

$$f'(x) = 3(x^2 - 2x + \tfrac{3}{4}) = 3(x - \tfrac{1}{2})(x - \tfrac{3}{2}).$$

The local maximum is at $x = \frac{1}{2}$, where

$$f(\tfrac{1}{2}) = \tfrac{1}{8} - \tfrac{3}{4} + \tfrac{9}{8} + c = \tfrac{1}{2} + c.$$

Choose $c = -1$. The equation is then

$$x^3 - 3x^2 + \tfrac{9}{4}x - 1 = 0 \qquad (a = -1,\ b = \tfrac{3}{4},\ c = -1),$$

and $a^2 = 1 > \frac{3}{4} = b > 0$, $a < 0$, $c < 0$, so $(*)$ holds.

Now count the roots. $f$ is increasing on $(-\infty, \frac{1}{2}]$, so there $f(x) \le f(\frac{1}{2}) = -\frac{1}{2} < 0$. On $[\frac{1}{2}, \frac{3}{2}]$, $f$ is decreasing, so it stays below $-\frac{1}{2}$. On $[\frac{3}{2}, \infty)$, $f$ increases from $f(\frac{3}{2}) = -1$ to $+\infty$, so it crosses the axis exactly once. Hence there is exactly one real root (it lies between 2 and 3, since $f(2) = -\frac{1}{2}$ and $f(3) = \frac{23}{4}$).

## Checks

- The cubic $(x-1)(x-2)(x-3) = x^3 - 6x^2 + 11x - 6$ has $a = -2$, $b = \frac{11}{3}$, $c = -6$, and indeed $a^2 = 4 > \frac{11}{3} > 0$, as part (ii) predicts.
- Reflecting the curve, $x \mapsto -x$, turns roots $\alpha, \beta, \gamma$ into $-\alpha, -\beta, -\gamma$ and changes the signs of $a$ and $c$ but not $b$. So $ab$ and $c$ both change sign, and the mirror image of part (i) says that when $c > 0$ either exactly one root is negative or all three are: exactly the two pictures we started from in part (iii).
- In part (iv) the only real root is $x \approx 2.178$. A computer check of thousands of random cubics with three distinct real roots also agreed with parts (i)–(iii).

## Common slips

- Trying to prove $a^2 > b$ from the signs of the roots alone. It comes from the turning points being real and distinct.
- Assuming that $(*)$ is *equivalent* to three positive roots. Part (iv) shows it is only necessary, not sufficient.
- In part (iv), choosing $c$ so that the local *minimum* is negative but the local maximum is still positive. That gives three roots, not one.

## Beyond STEP

The sign arguments here are the start of a general theory: Descartes' rule of signs bounds the number of positive roots by the number of sign changes in the coefficients, and Sturm's theorem counts real roots in an interval exactly. For a cubic, the condition missed in part (iv) is the sign of the discriminant, which is (up to a positive factor) the product $-f(p)f(q)$ of the values at the turning points. Allenby's *Rings, Fields and Groups* discusses polynomials and their roots at first-year level.

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